

LLMs

Language Models

Course Contents

Deep Learning

Transformer

LLMs

Context Engineering

Finetuning

Agents

The Bigger Picture

This Part: LLMs

- generative AI
- autoregressive text generation
- scaling laws
- LLM zoo

Generative vs Predictive/Discriminative Models

discriminative models:

predict conditional probability $P(Y|X)$

generative models:

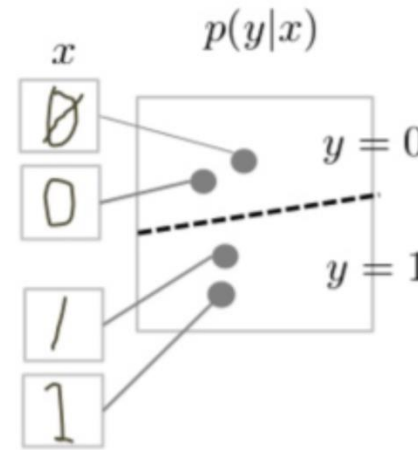
predict joint probability $P(Y, X)$

(or just $P(X)$ → unsupervised learning)

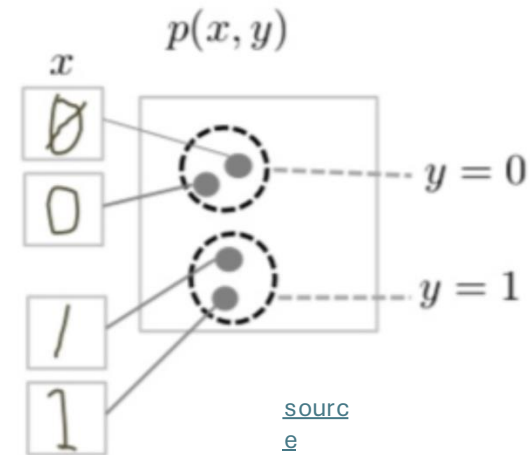
→ allow to generate new data samples

Generative models can be used for predictive tasks (Bayes theorem).
But predictive models are usually better at it.

discriminative model



generative model



[source](#)
[e](#)

task of generative models more difficult: need to model full data distribution rather than merely find patterns in inputs to distinguish outputs

Autoregressive Text Generation

LLMs (autoregressive transformers) generate one output token at a time (according to predicted probabilities)

then add it to the context for the prediction of the next token

→ resulting in a full text, sampled from predicted probabilities

How are LLMs Generative Models then?

at each step, take the sequence of previous tokens $x_{<t}$ (context) and predict a conditional probability distribution over the next token x_t :

$$P(x_t | x_{<t})$$

joint probability distribution over text sequences via chain rule of probability:

$$P(x_1, x_2, \dots, x_T) = \prod_{t=1}^T P(x_t | x_{<t})$$

→ no direct prediction of joint probability in one shot, but implicit calculation through product of conditionals

Prompting

feed information into decoder-only transformer via input prompt

→ attention to context

using all the knowledge acquired during internet-scale pretraining and stored in weights

kind of programmable neural network

→ a new paradigm: **one model for many tasks**

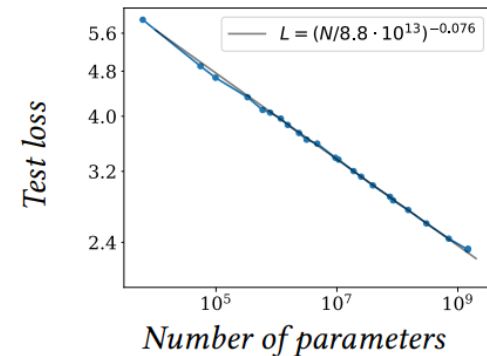
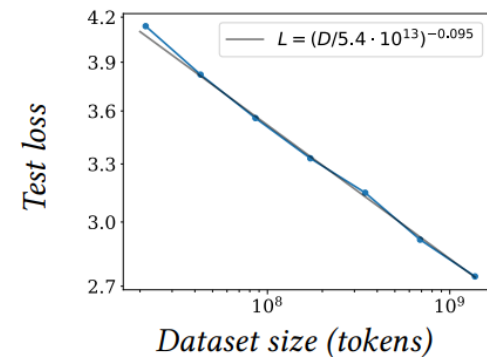
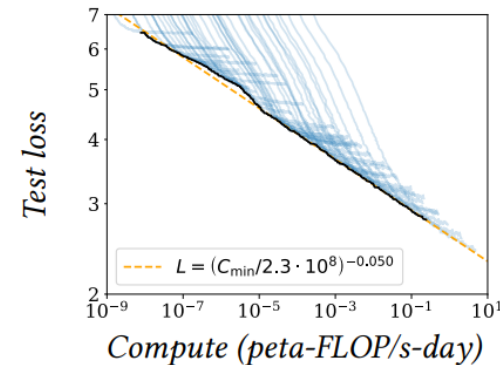
Size Matters: **LARGE** Language Models

scaling laws, Chinchilla: coupled performance power laws with model size, amount of training data, and compute

→ era of large-scale models

emergent abilities of LLMs:

- multi-task learning: perform new tasks at test time without task-specific training (via prompting)
- reasoning capabilities



Some LLM Numbers

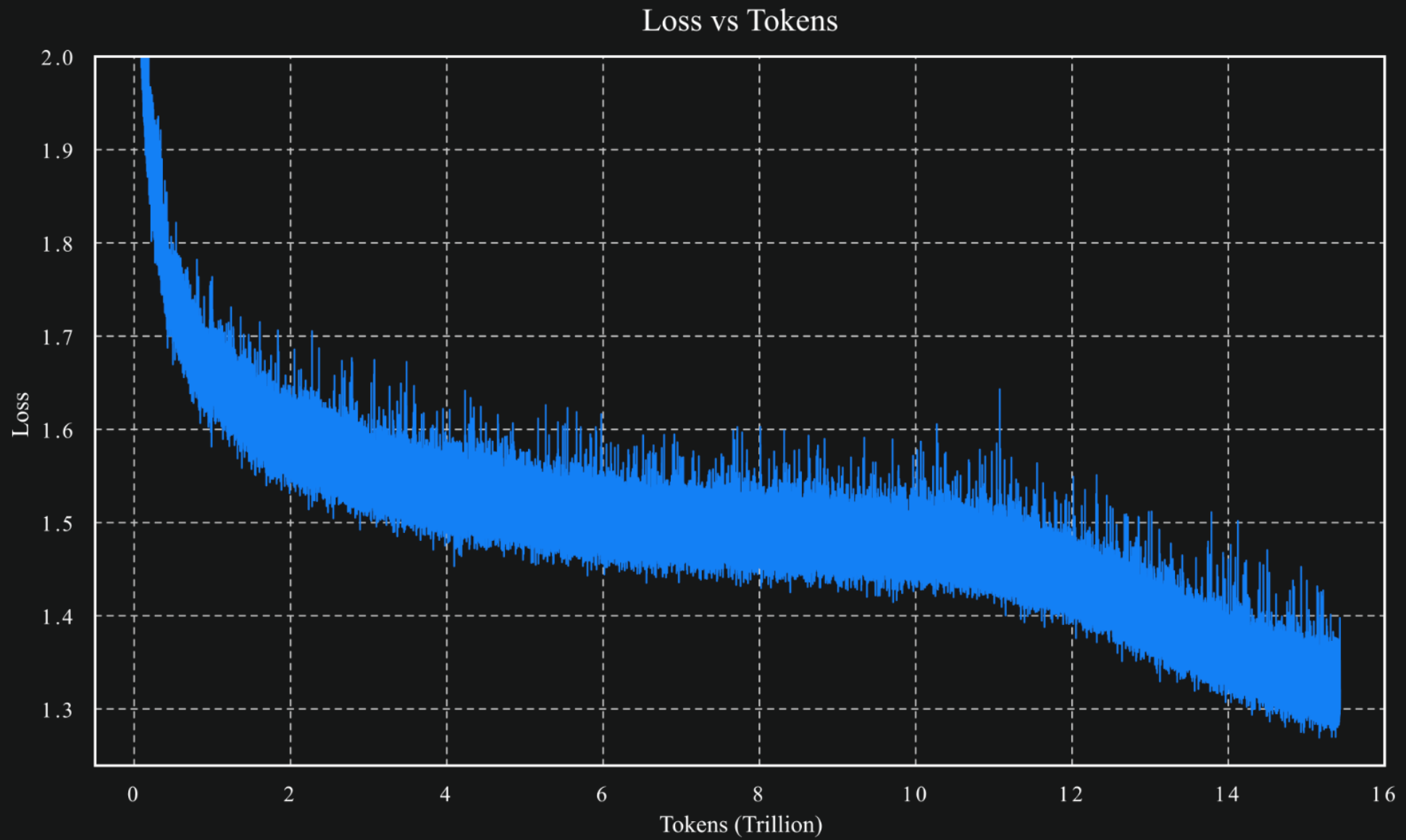
example [Kimi K2](#) (Mixture-of-Experts model):

- vocabulary size (tokens): 160K
- embedding dimension: 7,168
- context length (tokens): 128K
- total parameters: 1T
- training tokens: 15.5T
- number of layers: 61
- number of attention heads: 64
- number of experts: 384

as usual, comes in base and instruct variants

← → a lot of memorizing

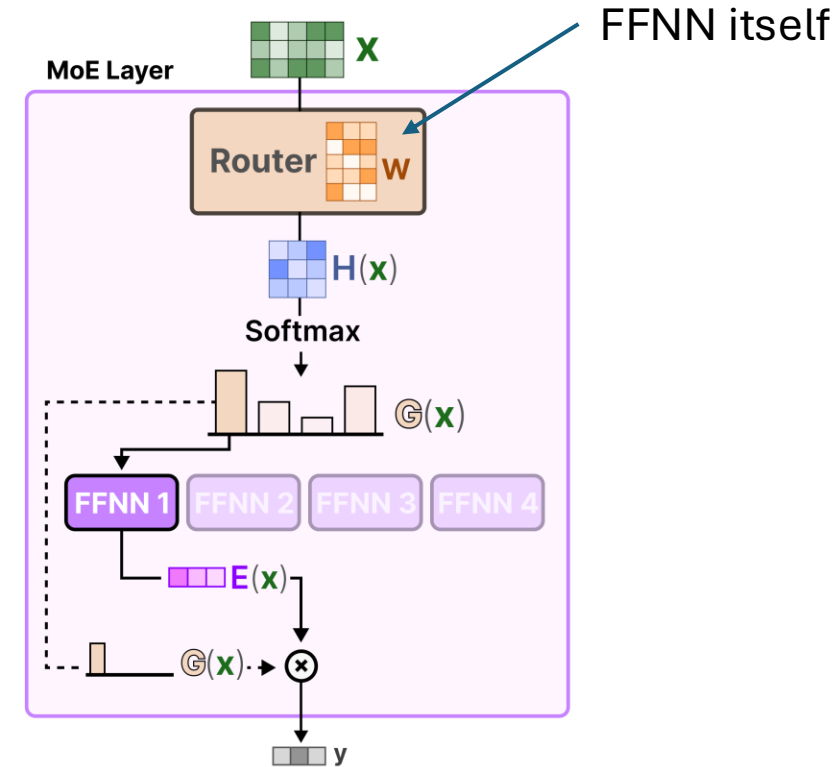
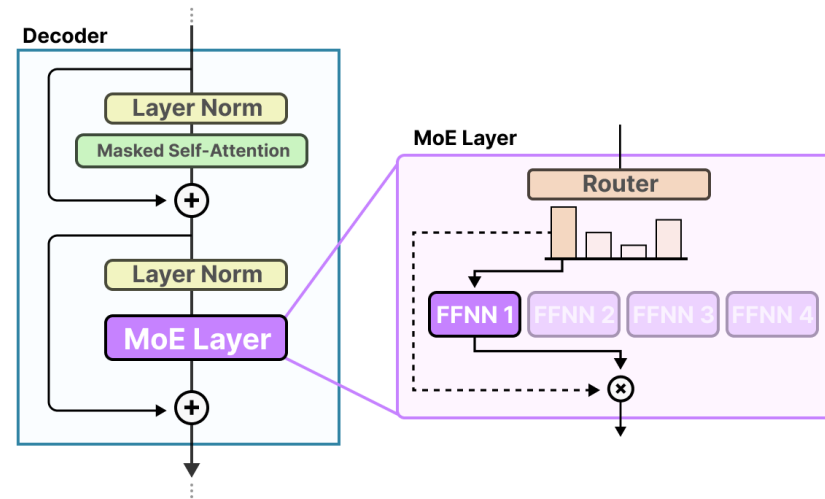
compression of the internet



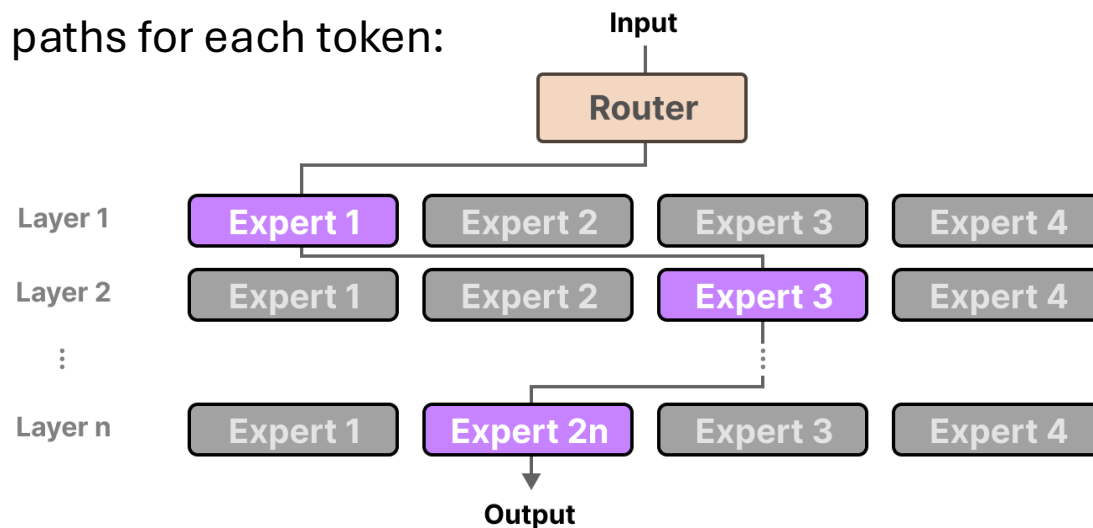
[source](#)

Mixture-of-Experts (MoE)

idea: replace big FFNN
with several smaller ones
and activate only few
→ less active parameters

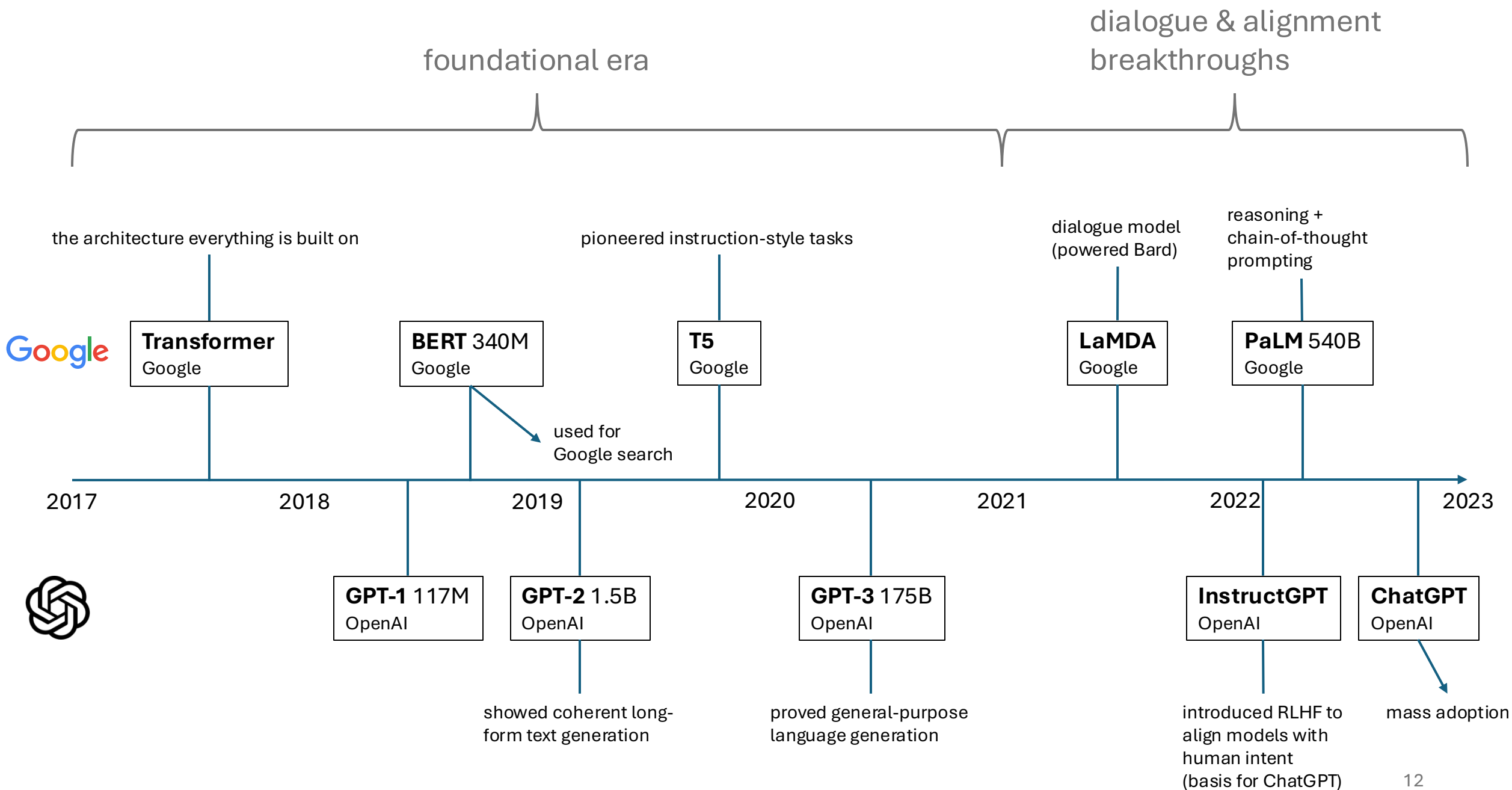


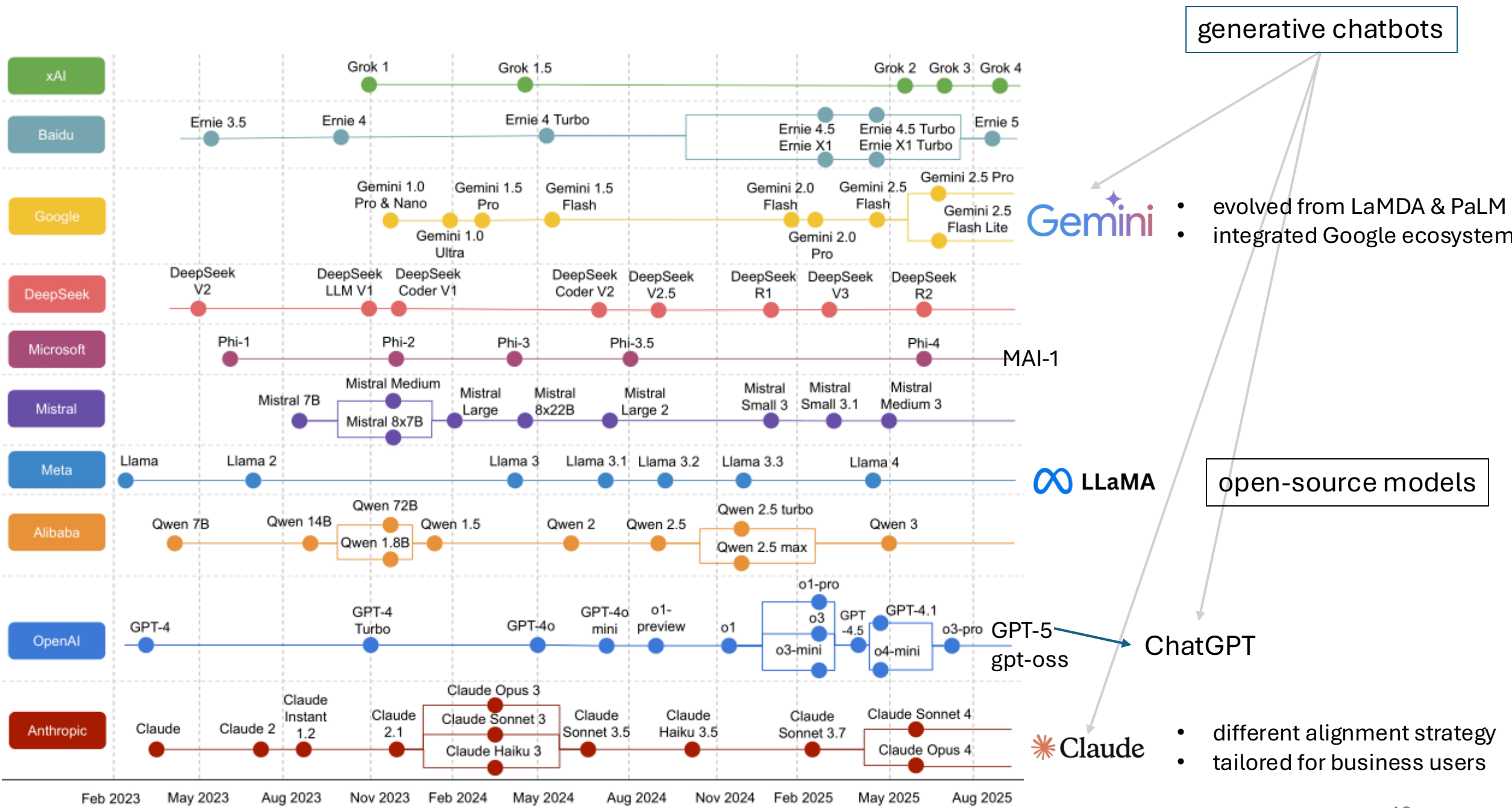
different paths for each token:



example (actually more abstract):







OpenAI's GPT Series

- [GPT-1](#): used for discriminative finetuning on specific tasks (summarization, translation, question-answering, ...) → BERT
- [GPT-2](#): zero- or few-shot learning → prompting
- [GPT-3](#): scaling (175 billion parameters) → much increased capabilities
- [GPT-4](#): more scaling ([capabilities](#))
- GPT-4o: multimodal model
- [o1](#): reasoning model
- GPT-5: automatic activation of reasoning model depending on the task

[gpt-oss](#)

LMarena

Rank (UB) ↑	Model ↓	Score ↓	95% CI (±) ↓	Votes ↓	Organization ↓	License ↓
1	 gemini-2.5-pro	1455	±5	41.731	Google	Proprietary
1	 claude-opus-4-1-20250805-thinking-16k	1451	±6	11.750	Anthropic	Proprietary
2	 o3-2025-04-16	1444	±4	43.898	OpenAI	Proprietary
2	 gpt-5-high	1442	±6	15.076	OpenAI	Proprietary
2	 chatgpt-4o-latest-20250326	1441	±4	36.426	OpenAI	Proprietary
3	 gpt-4.5-preview-2025-02-27	1439	±6	15.271	OpenAI	Proprietary
3	 claude-opus-4-1-20250805	1438	±6	18.341	Anthropic	Proprietary
5	 gpt-5-chat	1430	±6	11.808	OpenAI	Proprietary
6	 qwen3-max-preview	1428	±7	8.781	Alibaba	Proprietary
8	 grok-4-0709	1422	±5	21.446	xAI	Proprietary
8	 claude-opus-4-20250514-thinking-16k	1420	±5	27.351	Anthropic	Proprietary
8	 kimi-k2-0711-preview	1420	±5	23.527	Moonshot	Modified MIT
8	 qwen3-235b-a22b-instruct-2507	1420	±6	17.947	Alibaba	Apache 2.0
8	 deepseek-v3.1-thinking	1419	±8	8.226	DeepSeek	MIT
8	 deepseek-v3.1	1418	±7	9.822	DeepSeek	MIT
8	 deepseek-r1-0528	1418	±6	20.863	DeepSeek	MIT
8	 kimi-k2-0905-preview	1418	±11	2.918	Moonshot	Modified MIT
10	 glm-4.5	1411	±6	14.105	Z.ai	MIT

Another Trend: Small Language Models (SLM)

most powerful LLMs keep getting bigger (> 1 trillion parameters)

GPT, Claude, Gemini, ...

but slimmed-down (yet still strong) versions get smaller

[Gemma](#) (2B), [Phi-3](#) (3.8B), [Mistral 7B](#), [Llama3](#) (8B), ...

can run (inference) on laptops, smartphones, etc.

easier finetuning → specific tasks

Next: Context Engineering

To get the most out of LLMs, you need to prompt in the right way.

→ prompt engineering

Furthermore, the capabilities of LLMs can be enhanced by including additional information in the prompt.

→ context engineering